

5G FULL System Level

1 Architecture

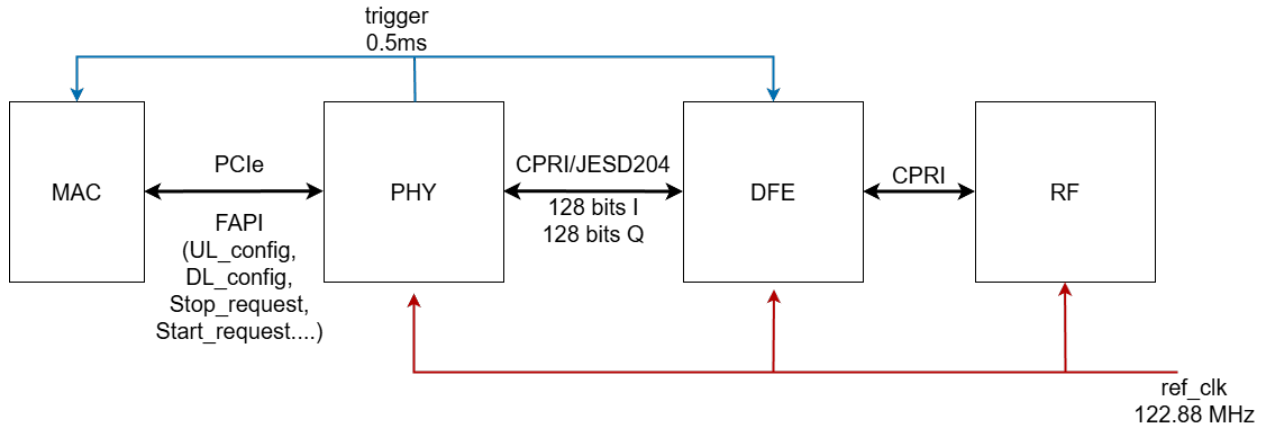


Figure 1: 5G full system level.

1.1 MAC

- Generates scheduling and control decisions and provides configuration commands to the PHY.

1.2 PHY

- Performs baseband physical-layer processing (e.g., PBCH, PDCCH, PDSCH, PUCCH, and PUSCH) and creates a **0.5 ms trigger** (i.e., **1 slot = 1 TTI**) for system synchronization.

1.3 DFE (Digital Front-End)

- Implements high-rate digital front-end processing and bridges the PHY baseband domain and the RF domain.

1.4 RF

- Implements radio-frequency transmission and reception.